ewater.services · Technical Request

MCP Tool Description Update Request

From: Robbie / Tapha AI Agent team · 18 July 2026

Why This Matters

The Tapha AI agent decides which tool to call, when, and with what inputs almost entirely by reading each tool's description. When descriptions only say what a tool returns but not where the inputs come from or what to call next, the agent guesses — and guesses are wrong.

Real example observed in production

User asked: "What water system does tag DBDA4ED2 use?"

The agent correctly called get_tag and got back system_id: 5, asset_id: 40. On the follow-up about the water system it invented "Chidilo / TIB Dodoma" instead of calling get_asset(40). The real answer — "Jafai Koto, Gambia" — was only found when the user was forced to ask "what asset is asset id 40?" explicitly.

Each tool description needs to answer three questions:

- When do I call this tool? (what user question triggers it)
- Where do my input parameters come from? (prior tool result, user message, etc.)
- What do I call next with my output? (which fields feed which other tools)

The Call Chain the Agent Must Follow

```
User mentions a tag ID
  !’ get_tag(tagId)
      % % %  primary_asset_id !’ get_asset(assetId)          !• water system name, tap type, country
      % % %  household_id     !’ get_household(householdId)  !• household name
User asks about performance / history / settings
  !’ get_asset_history / get_asset_flow_rate / get_asset_ewc_settings /
get_calibration_analysis
      % % %  assetId comes from get_tag.primary_asset_id or list_assets.id
```

`get_tag` UPDATE DESCRIPTION

REQUESTED:

Looks up an NFC water tag by its short tag ID (e.g. DBDA4ED2). Returns: primary_asset_id (numeric), primary_system_id (numeric), primary_country_id (numeric), household_id (UUID), sign-up date, last usage date, disbursement count, top-up count, credit balance, deleted status, blacklisted status.

When to call: Any time a user mentions a tag ID or asks about a specific NFC tag, household, or registered water user. This is always the starting point — call this first.

What to call next with the result:

- For the tap/dispenser name and water system name !’ get_asset(assetId = primary_asset_id)
- For the household name !’ get_household(householdId = household_id)

Never guess the household name, water system name, or country from the IDs returned — always resolve them via the appropriate follow-up tool call.

`get_asset (by ID)` MISSING — PLEASE ADD

& This tool does not appear in our tools/list response. It is the most critical missing tool for chaining.

REQUESTED:

Returns full details for one eWater asset given its numeric ID: asset name, asset type (e.g. CommunityTap), status (Active/Inactive), GPS coordinates, water system name, and country name.

When to call: When a user asks "what water system does this tag/household use?", "what tap is asset ID X?", or any question needing details about a specific asset. Call this after get_tag using primary_asset_id as the input.

Where assetId comes from: The primary_asset_id field returned by get_tag, or the id field from list_assets.

Never guess the water system name, country, or asset type — this tool returns the real values.

`get_household` MISSING — PLEASE ADD

& This tool does not appear in our tools/list response.

REQUESTED:

Returns the registered household name and details for a given household UUID.

When to call: After get_tag, when the user asks "who is this tag registered to?", "what is the household name?", or "whose account is this?" Pass the household_id UUID from get_tag.

Where householdId comes from: The household_id field returned by get_tag.

Never guess the household name — always call this tool when you have a household_id from a tag lookup.

`list_assets` UPDATE DESCRIPTION

CURRENT:

Lists all eWater assets (dispensers/taps) visible to the configured account, including id, name, type, status, location, and water system/country grouping.

REQUESTED:

Returns all eWater assets (taps/dispensers) visible to this account. Each item includes: id (numeric asset ID), name, asset type, status, GPS location, water system name, and country name.

When to call: When a user asks to list all taps, find an asset by name or location, count assets, or find an asset ID when only the name is known.

Do not call this if you already have a numeric asset ID (e.g. primary_asset_id from get_tag) — call get_asset directly instead.

What to call next: Use the id from any result as assetId for get_asset_history, get_asset_flow_rate, get_asset_ewc_settings, or get_calibration_analysis.

`get_asset_history` UPDATE DESCRIPTION

CURRENT:

Returns time-series history for one eWater asset over the last N days: tank height (water/chlorine), daily water inflow, battery voltage history + status, and dispense flow-rate history.

REQUESTED:

Returns time-series history for one eWater asset over the last N days (1–180, default 7). Includes: tank water and chlorine height over time, daily water inflow volume, battery voltage history and charge status, and dispense flow-rate history.

When to call: When asked about asset performance over time, water levels, tank fill history, battery health trends, or historical flow rates for a specific tap or dispenser.

Where assetId comes from: The primary_asset_id field from get_tag, the id field from list_assets, or a numeric asset ID mentioned in the conversation. Never guess the assetId.

`get_asset_ewc_settings` UPDATE DESCRIPTION

CURRENT:

Returns the electronic water controller (EWC) device settings for one asset: flow conversion factor (FCF), litres conversion factor (LCF), currency conversion (FX), preload charge, price of water, and other device configuration values.

REQUESTED:

Returns the EWC (electronic water controller) configuration for one eWater asset: flow conversion factor (FCF), litres conversion factor (LCF), currency/FX settings, preload charge, price per litre, and other device configuration values.

When to call: When asked about pricing, how much water a credit buys, device configuration, preload settings, or calibration factors for a specific tap or dispenser.

Where assetId comes from: The primary_asset_id field from get_tag, the id field from list_assets, or a numeric asset ID mentioned in the conversation.

`get_asset_flow_rate` UPDATE DESCRIPTION

CURRENT:

Returns the most recent dispense flow rate (litres/minute) for one asset, derived from the last 24 hours of device logs.

REQUESTED:

Returns the most recent dispense flow rate in litres/minute for one eWater asset, calculated from the last 24 hours of device logs.

When to call: When asked "is the tap flowing properly?", "what is the current flow rate?", "how fast is water dispensing?", or any question about live dispensing performance for a specific asset.

Where assetId comes from: The primary_asset_id field from get_tag, the id field from list_assets, or a numeric asset ID mentioned in the conversation.

`get_calibration_analysis` UPDATE DESCRIPTION

CURRENT:

Runs a calibration and NRW gap analysis for one asset...

REQUESTED:

Runs a calibration and non-revenue-water (NRW) gap analysis for one eWater asset over N days (1–180, default 30). Compares the configured litres-conversion factor (LCF) against measured dispense events to detect meter drift. Compares configured preload against actual 'no credit' event data. Returns: LCF gap %, preload gap, and a plain-language summary of whether the meter is over-reading or under-reading.

When to call: When asked about NRW, meter drift, calibration accuracy, revenue loss, or whether LCF or preload settings are correct for a specific tap.

Where assetId comes from: The primary_asset_id field from get_tag, the id field from list_assets, or a numeric asset ID mentioned in the conversation.

Summary of Action Items

`get_tag`	Update description — add 'what to call next' section
`get_asset (by ID)`	ADD/EXPOSE this tool — currently missing from tools/list
`get_household`	ADD/EXPOSE this tool — currently missing from tools/list
`list_assets`	Update description — add when NOT to call it + what next
`get_asset_history`	Update description — add where assetId comes from
`get_asset_ewc_settings`	Update description — add where assetId comes from
`get_asset_flow_rate`	Update description — add where assetId comes from
`get_calibration_analysis`	Update description — add where assetId comes from